Khush Jain

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EDUCATION

University of Wisconsin–Madison — Madison, WI
B.S. in Computer Science & Data Science; Minor in Geospatial Data Science — Sep 2023 – May 2027
GPA: 3.6/4.0 (Dean's List)

EXPERIENCE

Undergraduate Researcher – Multimodal Foundation Models — Sep 2026 – Present
GeoDS Lab, University of Wisconsin–Madison; Advisor: Prof. Song Gao — Madison, WI

- Developing PyTorch pipelines for multimodal representation learning across spatial and temporal data; benchmarking cross-modal alignment for location-based recommendation and mobility analysis in GeoAI systems.

Software Engineering & Geospatial Computing Intern — Aug 2026 – Present
MultiSet AI — Madison, WI

- Engineered Python REST APIs using OGC GeoPose for 6-DoF metadata validation and coordinate transforms, reducing cross-platform pose-conversion latency by **35%**.
- Developed point-cloud and computer-vision pipelines to detect low-feature-density regions and predict localization drift, improving 3D localization accuracy by **25%**.

Software Engineering & DevOps Intern — Jun 2026 – Aug 2026
Mass General Brigham Digital Innovation — Boston, MA

- Extended and deployed an existing GitHub Copilot analytics platform for **561 seats** using TypeScript, Vue/Nuxt, Node.js/Nitro, GitHub REST APIs, and PostgreSQL.
- Developed **10+ analytics modules** for executive KPIs, usage trends, cost attribution, and license efficiency across 244 developers and 144K+ chat interactions.
- Implemented GitHub OAuth, server-side RBAC, backend filtering, and automated tests; deployed Docker containers to OpenShift and flagged **267 unused seats** (up to **$8K/month** in potential waste).

Cloud & Artificial Intelligence Intern — May 2025 – Aug 2025
Mass General Brigham Digital Innovation — Boston, MA

- Developed an assistant using NLP and semantic search over **2,000+ clinical workflow and policy documents**, reducing repetitive support tickets by **28%**.
- Integrated intent classification and named-entity recognition with low-confidence escalation, achieving **91% routing accuracy** across 15 inquiry categories.
- Implemented a React/FastAPI interface delivering cited LLM answers in **under 2.5 seconds** to 1,500+ pilot users.

PROJECTS

DataForge | *Python, gRPC, Spark, Kafka, Cassandra, HDFS* — Nov 2025

- Engineered an end-to-end distributed data platform spanning SQL-to-HDFS ETL, Spark analytics, Cassandra storage, and Kafka streaming; processed **447K+ loan records**.
- Implemented an 8-threaded gRPC service within a 512 MB memory limit, replication and checkpoint recovery, and a 4-partition Kafka pipeline with crash-safe Parquet storage.

TaskPulse | *Python, Redis, Asyncio, REST APIs, Docker* — Oct 2026

- Engineered a multithreaded task queue and caching engine using Redis Pub/Sub for distributed coordination, processing **15,000+ req/sec** with **p99 latency below 5 ms**.
- Implemented sliding-window rate limiting and write-through caching in Redis, reducing backend database load by **65%** under high-concurrency synthetic load.

TerraMind | *Python, React, PostGIS, GeoPandas, Leaflet* — May 2026

- Developed a React/Leaflet map and Python APIs for PostGIS spatial search and optimized large-dataset rendering.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, TypeScript, JavaScript, C#, C, SQL, Node.js, React, Vue/Nuxt, FastAPI
AI & ML: PyTorch, Transformers, RAG, FAISS, MCP, LoRA, Quantization, Prompt Engineering
Cloud & Systems: Azure, Docker, Kubernetes, OpenShift, Terraform, Linux, Git, CI/CD, Bash, Bicep
Data & Tools: Redis, PostgreSQL, MySQL, Spark, Kafka, Cassandra, HDFS, Pandas, GeoPandas, ArcGIS Pro